

SQL Lecture Notes (5/20/2026)

Data Query Language:

- i) Select (*) from considers all columns from the table.
- ii) Select DISTINCT is used to return only unique values from one or more columns.
- iii) Select COUNT (*) is used to count the total number of rows in a table.
- iv) Select count (distinct column_name) from TABLE counts how many unique (non-NULL) values exist in a specific column.
- v) LIMIT is used to restrict how many rows are returned.
- vi) ORDER BY is used to sort results.
- vii) AND, OR are logical operators used in a WHERE clause to filter rows.
- viii) AND Returns rows only if all conditions are true. OR Returns rows if at least one condition is true.

LIKE Operators:

Like operator is used for pattern matching in text. It works with two main wildcards:

- i) Percentile (%) - Represents zero, one or many characters.
- ii) Underscore - Represents exactly one character.

Between Operator:

BETWEEN is used to filter values within a range (inclusive).

Group By and Having:

- i) GROUP BY is used to group rows that have the same values in one or more columns, usually to perform aggregate functions like COUNT, SUM, AVG, MAX, MIN.
- ii) HAVING is used to filter grouped results (after GROUP BY).

Order of execution:

From (table) – Join – Where – Group By – Having – Order By – Limit.

Built in functions:

Padding is used to put extra space.

- Strings → Text values stored in columns (e.g., 'Hello').
- SUBSTRING(string, start, length) → Extracts part of a string from a given position for a given length.
- Substring range → Extracts characters between positions using SUBSTRING(column, start, length).
- CONCAT(string1, string2, ...) → Joins multiple strings together into one.
- LENGTH(string) → Returns the number of characters in a string.
- SUBSTRING with LOCATE → Extracts text based on a position found using LOCATE(substring, string).
- SUBSTRING_INDEX(string, delimiter, count) → Returns part of a string before/after a delimiter based on count.
- UPPER(string) → Converts all characters in a string to uppercase.
- LOWER(string) → Converts all characters in a string to lowercase.
- LEFT(string, n) → Returns the first n characters from the left side of a string.
- RIGHT(string, n) → Returns the last n characters from a string.

We use multiple built in functions to get the output.

Case Statement:

Conditional logic in SQL that returns different values based on conditions (like IF-ELSE).

- REPLACE(string, old, new) → Replaces all occurrences of a specified substring with another substring in a string.
- REGEXP (Regular Expression) → Pattern matching operator used to search, match, or filter strings based on complex patterns.